

On Pumping Constants and Smallest Grammars for Context-Free Languages, Part II—Greibach Normal Form

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Abstract

We continue our research, started in [H. Gruber and M. Holzer. On pumping constants and smallest grammars for context-free languages, LNCS 15840, pp. 54–68, 2025], on the relation between context-free grammar size and minimal pumping constants w.r.t. Bar-Hillel’s pumping lemma, now considering the case when the language is described by a grammar in Greibach normal form.

1 Introduction

Understanding the structural complexity of grammars that generate context-free languages is a central topic in formal language theory. Among the many quantitative measures of grammatical succinctness, the *Smallest Grammar Problem* (SGP), see, e.g., [5, 10]—which seeks to determine, for a given string or language, the smallest context-free grammar that generates it—provides a natural and meaningful way to

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assess the expressive efficiency of different grammatical restrictions. The SGP lies at the intersection of formal language theory, data compression, and computational complexity, offering a fundamental measure of how efficiently a structure or sequence can be described. In its essence, it formalizes the intuitive notion of succinctness, quantifying, through the framework of grammatical generation, how compactly information can be represented without loss. Beyond its immediate computational relevance, the SGP offers rich mathematical structure. The optimal grammar size for language classes often follows nontrivial recurrences and growth patterns, revealing intricate dependencies on parameters such as the grammar’s form, alphabet, or pumping constant. The present authors studied in [7] the relationship between the pumping constant and the size of the smallest grammar for a given context-free language. There it was shown that the optimal grammars generate a single unary word, and are based on recursively tripling the length of the string. The function $\rho(n)$ describing the optimal bound is the Moon-Moser [12] sequence, with $\rho(n) = 3^{n/3}$, if n is divisible by 3, and with similar formulas for each of the other two remainders modulo 3. That study also revealed a non-trivial relationship to the *Keystroke Problem* [15], a recreational mathematical puzzle. The problem asks, given a positive integer n , for the maximum amount of characters that can be produced using n keystroke actions—the actions include keyboard shortcuts for copy-and-paste *manoeuvres*.

In this paper we continue our research on the relationship between the pumping constant of a context-free language and the size of the smallest grammar generating it. Here, we consider grammars in Greibach normal form [6]. Our results uncover a remarkably intricate behavior of the optimal grammar size function, denoted by $\gamma(n)$. Like for unrestricted context-free grammars, the worst-case scenario occurs for singleton unary languages, where the corresponding optimal grammars exhibit a recursively self-similar structure: each step effectively quadruples the length of the generated string while producing exactly one terminal symbol per rule. Interestingly, this quadrupling pattern emerges only in the asymptotic regime. For smaller input sizes (up to $n = 20$), the optimal construction instead relies on tripling or even doubling, indicating a transient phase before the asymptotic behavior stabilizes. Furthermore, the growth of $\gamma(n)$ is depending on the remainder of $n \bmod 5$, which reveals a subtle periodic structure underlying the recurrence, a bit similar to the case of unrestricted context-free grammars. The recurrence defining γ involves a maximum operator. This poses significant analytical challenges, as it combines discrete optimization with functional recurrence. Unlike linear, or convex recurrences, where standard techniques such as generating functions, characteristic equations, and asymptotic expansions apply, max-type recurrences encode a non-convex optimization process at each step. This step entails the full history of the recurrence. Solving these recurrences requires locating the index of the maximizing term and characterizing how it evolves with n . This typically entails a piecewise analysis, since the maximizing index may vary as n increases, leading to periodic growth patterns in the solution. In particular, max recurrences often produce integer sequences with irregular local behavior but a tame asymptotic behavior, making them simultaneously rich in analytical structure and sometimes resistant to closed-form resolution. For more background on integer recurrences, see, e.g., [8, 3, 13].

In this context, the present paper contributes to a deeper understanding of the smallest grammar problem by analyzing the relation between grammatical structure, succinctness, and asymptotic behavior. By studying how minimal grammar size evolves under specific normal forms and recursive constructions, we reveal new insights into the inherent trade-offs between expressiveness and compactness—a theme that lies at the heart of formal language theory and computational information representation.

2 Preliminaries

We recall a few notions regarding normal forms of context-free grammars. For further background, see [1]. Let $G = (N, T, P, S)$ be a *context-free grammar* (CFG), where N and T are disjoint alphabets of *nonterminals* and *terminals*, respectively, $S \in N$ is the *axiom*, and P is a finite set of *productions* of the form $A \rightarrow \alpha$, where $A \in N$ and $\alpha \in (N \cup T)^*$. As usual, the transitive closure of the derivation relation $\Rightarrow_G$ is written as $\Rightarrow_G^*$. If there is no danger of confusion, we simply write $\Rightarrow$ ($\Rightarrow^*$, respectively) instead of $\Rightarrow_G$ ($\Rightarrow_G^*$, respectively). The *language generated* by G is defined as

$$L(G) = \{ w \in T^* \mid S \Rightarrow_G^* w \}.$$

In this paper, the *size* of a context-free grammar is understood as the sum of the length of the right-hand side of all productions of G ; we thus follow the convention used in literature on the smallest grammar problem, see, e.g., [5].

For every context-free language the Bar-Hillel lemma [4], also known as pumping lemma, applies, which reads as follows:

Lemma 1 Let L be a context-free language over Σ . Then, there is a constant p (depending on L) such that the following holds: If $z \in L$ and $|z| \geq p$, then there are words $u, v, w, x, y \in \Sigma^*$ such that $z = uvwxy$, $|vx| \geq 1$, $|vwx| \leq p$, and $uv^t wx^t y \in L$ for $t \geq 0$ —it is then said that v and x can be (simultaneously) *pumped* in z .

For a context-free language L , let $\text{mpcf}(L)$ denote the minimal number p satisfying the conditions of Lemma 1. For instance, consider the context-free grammar $G_k = (\{S\}, \{a, b\}, P_k, S)$ with the set of productions P_k containing exactly the rules

$$S \rightarrow a^{k-1} S b \mid \varepsilon,$$

where ε refers to the empty word of length 0. Obviously, the context-free grammar G_k of size $k + 1$ generates the language $\{a^{n(k-1)} b^n \mid n \geq 0\}$. By inspection, the sub-words a^{k-1} and b can be pumped in any word of $L(G_k)$ but any shorter word cannot since otherwise the number of a 's and b 's is no longer well correlated anymore. Hence, the minimal pumping constant w.r.t. Lemma 1 is equal to k . Thus, $\text{mpcf}(T_k) = k$, where T_k is equal to the language $L(G_k)$.

Recently, a tight bound on the minimal context-free pumping constant in terms of the sum of the lengths of the right-hand sides of the grammar was shown in [7]. For this, the function ρ was needed, which became known as the maximum product

function w.r.t. a partition of the number n [9], that is, the value $\rho(n)$ is obtained by maximizing the product $\prod_{i=0}^h k_i$ of positive integers k_i subject to the condition $\sum_{i=0}^h k_i = n$, for $1 \leq k_i \leq n$ with $0 \leq i \leq h$. This function became also known in graph theory as the Moon-Moser bound [12], see also [11].

Definition 1 For a positive integer n , let the function $\rho(n)$ be defined as follows:

$$\rho(n) = \begin{cases} 0, & \text{for } n = 0 \\ 1, & \text{for } n = 1 \\ 3^{\frac{n}{3}}, & \text{for } n \equiv 0 \pmod{3} \text{ and } n \geq 1 \\ 4 \cdot 3^{\frac{n-4}{3}}, & \text{for } n \equiv 1 \pmod{3} \text{ and } n \geq 4 \\ 2 \cdot 3^{\frac{n-2}{3}}, & \text{for } n \equiv 2 \pmod{3}. \end{cases}$$

Then the following holds:

Theorem 2 ([7]) *Let L be generated by the context-free grammar $G = (N, T, P, S)$, where L is not empty. Then $\text{mpcf}(L) \leq \rho(m_G) + 1$, where $m_G := \sum_{(A \rightarrow \alpha) \in P} |\alpha|$ and ρ is the function from Definition 1. Furthermore, for every $m \geq 0$, there exists a context-free grammar G_m witnessing that this bound is tight.*

Other normal forms such as, e.g., Greibach normal form, or variants thereof, were not investigated so far, up to our knowledge. A context-free grammar $G = (N, T, P, S)$ is in *Greibach normal form* [6] if each rule is of the forms

$$\begin{aligned} A &\rightarrow aB_1B_2 \cdots B_n \\ A &\rightarrow a, \end{aligned}$$

where $A, B_1, B_2, \dots, B_n \in N$ and $a \in T$. It is worth mentioning that the Greibach normal form finds immediate application in parser construction [2], since it allows to overcome the problem on left-recursive nonterminals. A grammar is said to be in *m-standard form* if G is in Greibach normal form, as above, and $n \leq m$, see [1]. Thus, a grammar is in Greibach normal form if and only if it is in *m-standard form* for some m . While the languages generated by grammars in 1-standard form are obviously the regular languages, it is well known that every context-free language admits a context-free grammar in 2-standard form [14, 1].

3 Pumping Constants and 2-Standard Form

For determining the minimal pumping constant, it makes sense to look at the length of a longest word that cannot be pumped. As in [7], good candidates for grammars that attain extremal values are acyclic context-free grammars that generate a single word; the argument will be made rigorous later. As a warm-up, let us look at grammars that generate a single word and are in 2-standard form. Let us first look at an example.

Example 1 Let $G = (\{A_1, A_2, \dots, A_8\}, \{a\}, P, A_1)$ be the context-free grammar with the following productions:

$$\begin{array}{ll} A_1 \rightarrow aA_2A_2 & A_5 \rightarrow aA_6A_6 \\ A_2 \rightarrow aA_3A_3 & A_6 \rightarrow aA_7A_7 \\ A_3 \rightarrow aA_4A_4 & A_7 \rightarrow aA_8 \\ A_4 \rightarrow aA_5A_5 & A_8 \rightarrow a \end{array}$$

The grammar is acyclic, has size 21, and generates the single word a^{191} . The example shows that the minimal pumping constant w.r.t. Bar-Hillel's pumping lemma is at least 192.

In general, what is the longest word that can be generated by some acyclic grammar of size n in 2-standard form? Disregarding degenerate cases, the start symbol of such an acyclic grammar in 2-standard form appears as left-hand side in at least one production, whose right-hand side can have length at most 3. If we use the first nonterminal at the right hand side of this production as the new start symbol, and remove all productions that involve the previous start symbol, we obtain a smaller acyclic grammar in 2-standard form. This gives rise to a relatively simple recurrence for an upper bound.

Lemma 3 For a positive integer n , let the function $\gamma_2(n)$ be given by $\gamma_2(1) = 1$, $\gamma_2(2) = 1$, $\gamma_2(3) = 2$, and the recurrence

$$\gamma_2(n) = \max\{\gamma_2(n-2) + 1, 2 \cdot \gamma_2(n-3) + 1\} \quad \text{for } n \geq 4.$$

Then

$$\gamma_2(n) = \begin{cases} 1, & \text{if } n = 1, 2, \\ 3 \cdot 2^{k-1} - 1, & \text{if } n = 3k, \text{ for some } k \geq 1, \\ 2^{k+1} - 1, & \text{if } n = 3k + 1 \text{ or } n = 3k + 2, \text{ for some } k \geq 1. \end{cases}$$

Proof First observe, that the values for $\gamma_2(n)$ as described by the recurrence and the closed formula are identical for $n = 1, 2, 3$. For the remaining values, we prove the statement by induction on k . Let $k = 1$. Then we have to verify the values of γ_2 for $n = 3, 4, 5$, where only for $n = 3$ and $n = 4$ the recurrence applies. We have

$$\gamma_2(3) = 2 = 3 - 1 = 3 \cdot 2^0 - 1 = 3 \cdot 2^{1-1} - 1.$$

For $\gamma_2(4)$ we find

$$\begin{aligned} \gamma_2(4) &= \max\{\gamma_2(2) + 1, 2 \cdot \gamma_2(1) + 1\} \\ &= \max\{1 + 1, 2 \cdot 1 + 1\} = 3 = 4 - 1 = 2^2 - 1 = 2^{1+1} - 1 \end{aligned}$$

and

$$\begin{aligned} \gamma_2(5) &= \max\{\gamma_2(3) + 1, 2 \cdot \gamma_2(2) + 1\} \\ &= \max\{2 + 1, 2 \cdot 1 + 1\} = 3 = 4 - 1 = 2^2 - 1 = 2^{1+1} - 1. \end{aligned}$$

This shows the induction base case. Now assume that the induction hypothesis holds for $k-1$, that is, for the values of γ_2 for $n = 3k-3, 3k-2, 3k-1$. Next we do the induction step from $k-1$ to k . Assume $k \geq 2$. Consider $\gamma_2(3k)$ first. Then we have

$$\gamma_2(3k) = \max\{\gamma_2(3k-2) + 1, 2 \cdot \gamma_2(3k-3) + 1\}$$

$$\begin{aligned}
&= \max\{\gamma_2(3(k-1)+1)+1, 2 \cdot \gamma_2(3(k-1))+1\} \\
&= \max\{2^{(k-1)+1}-1+1, 2 \cdot (3 \cdot 2^{(k-1)-1}-1)+1\} \\
&= \max\{2^k, 3 \cdot 2^{k-1}-1\} \\
&= 3 \cdot 2^{k-1}-1.
\end{aligned}$$

Analogously we get

$$\begin{aligned}
\gamma_2(3k+1) &= \max\{\gamma_2(3k-1)+1, 2 \cdot \gamma_2(3k-2)+1\} \\
&= \max\{\gamma_2(3(k-1)+2)+1, 2 \cdot \gamma_2(3(k-1)+1)+1\} \\
&= \max\{2^{k-1}-1+1, 2 \cdot (2^k-1)+1\} \\
&= \max\{2^{k-1}, 2^{k+1}-1\} \\
&= 2^{k+1}-1
\end{aligned}$$

and

$$\begin{aligned}
\gamma_2(3k+2) &= \max\{\gamma_2(3k)+1, 2 \cdot \gamma_2(3k-1)+1\} \\
&= \max\{\gamma_2(3k)+1, 2 \cdot \gamma_2(3(k-1)+2)+1\} \\
&= \max\{(3 \cdot 2^{k-1}-1)+1, 2 \cdot (2^k-1)+1\} \\
&= \max\{3 \cdot 2^{k-1}, 2^{k+1}-1\} \\
&= 2^{k+1}-1,
\end{aligned}$$

as desired. This proves the stated claim. $\square$

The first few values for γ_2 are given in Table 1. It is worth mentioning that $\gamma_2(n)$

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
$\gamma_2(n)$	1	1	2	3	3	5	7	7	11	15	15	23	31	31	47	63	63	95

Table 1 Values of $\gamma_2(n)$ for $n = 1, 2, \dots, 18$.

for odd n is equal to

$$1, 2, 3, 7, 11, 15, 31, 47, 63, 127, 191, 255, 511, 767, 1023, \dots$$

and appears in the “Online Encyclopedia of Integer Sequences” (OEIS) under sequence number [A180516](#) and obeys the form $i \cdot 4^j - 1$, for $i = 1, 2, 3$ and $j \geq 0$, excluding the first term for $i = 1$ and $j = 0$.

The function γ_2 gives an upper bound on the size of the longest word in an acyclic context-free grammar in 2-standard form that cannot be pumped; but the recurrence of γ_2 guides us to construct a small context-free grammar in 2-standard form, which describes a word of the largest possible length. We briefly explain how to construct such a grammar.

Example 2 Let $n = 21$. A close inspection reveals that $\gamma_2(21) = 2 \cdot \gamma_2(18) + 1$, $\gamma_2(18) = 2 \cdot \gamma_2(15) + 1$, $\gamma_2(15) = 2 \cdot \gamma_2(12) + 1$, $\gamma_2(12) = 2 \cdot \gamma_2(9) + 1$, $\gamma_2(9) = 2 \cdot \gamma_2(6) + 1$, $\gamma_2(6) = 2 \cdot \gamma_2(3) + 1$,

$\gamma_2(3) = \gamma_2(1) + 1$, and $\gamma_2(1) = 1$. The recursive computation indicates how to construct an acyclic context-free grammar G of size 21 in 2-standard form, where A_1 is the axiom:

$$\begin{array}{ll} A_1 \rightarrow aA_2A_2 & A_5 \rightarrow aA_6A_6 \\ A_2 \rightarrow aA_3A_3 & A_6 \rightarrow aA_7A_7 \\ A_3 \rightarrow aA_4A_4 & A_7 \rightarrow aA_8 \\ A_4 \rightarrow aA_5A_5 & A_8 \rightarrow a \end{array}$$

Not surprisingly, this is the same grammar as in Example 1. Recall that G is of size 21 and by our previous considerations, we have that $a^{\gamma_2(21)} = a^{191}$ belongs to $L(G)$; in fact $L(G) = \{a^{191}\}$. Thus, the minimal pumping constant w.r.t. Bar-Hillel's pumping lemma is $191 + 1 = 192$.

In general, if L is a context-free language that admits a grammar in 2-standard form of size at most m , then $\text{mpcf}(L) \leq \gamma_2(m) + 1$, and this bound is tight. Let us now turn to the general case, where we also present the corresponding argument in a rigorous manner. The reader will then be able to reconstruct the full proof for 2-standard form with ease.

4 Pumping Constants and Greibach Normal Form

For a grammar in (unrestricted) Greibach normal form of size m , the length of the right-hand sides of productions may vary between 1 and $m - 2$. This motivates the following recurrence:

Definition 2 Let n be a positive integer. Then the function $\gamma(n)$ is given by the recurrence

$$\gamma(n) = \max_{1 \leq k \leq n-2} \{k \cdot \gamma(n-k-1) + 1\}, \quad \text{for } n \geq 3,$$

with base values $\gamma(1) = \gamma(2) = 1$.

The following theorem justifies this definition; the proof strategy is similar as in the case of unrestricted grammars [7, Thm. 5].

Theorem 4 Let L be generated by the context-free grammar $G = (N, T, P, S)$ in Greibach normal form, where L is not empty. Then

$$\text{mpcf}(L) \leq \gamma(m_G) + 1,$$

where $m_G := \sum_{(A \rightarrow \alpha) \in P} |\alpha|$ and γ is the recursively defined function from Definition 2. Furthermore, for every $m \geq 1$, there exists a context-free grammar G_m witnessing that this bound is tight.

Proof We prove the upper bound using the following statement: let w be a word in $L(G)$; if the word w cannot be pumped, then $|w| \leq \gamma(m_G)$.

We derive this statement by induction on the parameter m_G . For the base case of the induction we have a grammar G with $m_G = 1$, and the only production of G is of the form

$S \rightarrow a$ with $a \in \Sigma$. We have $\gamma(1) = 1$, thus establishing the base case of the induction. For the induction step, no grammar in Greibach normal form has $m_G = 2$, so assume $m_G \geq 3$. Further, assume that the claim holds for all grammars G' with $m_{G'} < m_G$. We consider two cases. Consider a parse tree T for w :

Case 1. There is a path in T from the root to some leaf labeled with a terminal symbol, such that the start symbol S appears at least twice.

The start symbol appears as a label at the root of the parse tree. There is also a proper subtree of T , whose root is labeled by S . This subtree is the parse tree of a word y according to the grammar G , and y is a contiguous subword of w . Also, y is a proper subword of w , since G is in Greibach normal form. Thus we can write $w = xyz$ with $|x| > 0$ and $|y| > 0$, and $S \Rightarrow^* xSz$ as well as $S \Rightarrow^* y$. Hence, we have $x^i y z^i \in L(G)$, for all $i \geq 0$, and w can be pumped. We have derived a contradiction in this case.

Case 2. On every path in T from the root to some leaf labeled with a terminal symbol, the start symbol S appears only once.

There is a production $p = (S \rightarrow aB_1B_2 \dots B_k)$ that corresponds to the root of the parse tree and its children, with $a \in \Sigma$ and $B_i \in V$ for $1 \leq i \leq k$ with $k \geq 1$. Notice that we must have $k \geq 1$, since the depth of the parse tree is at least 2 in this case; also, we must have $k \leq m_G - 2$ since G is in Greibach normal form. For $1 \leq i \leq k$, the variable B_i is at the root of some subtree, whose leaves spell out a proper subword x_i of w . Together with the initial terminal symbol a , these subwords make up the word $w = ax_1x_2 \dots x_k$. Notice that neither of the subtrees uses the variable S , and also not p . The subtree with root B_i is a parse tree of the context-free grammar

$$G_i = (N \setminus \{S\}, T, P \setminus \{p\}, B_i).$$

We thus obtain $m_{G_i} \leq m_G - k - 1$.

Recall that $w = ax_1x_2 \dots x_k$, and by the induction hypothesis, $|x_i| \leq \gamma(m_{G_i})$. With γ being nondecreasing, summing up yields

$$\begin{aligned} |w| &= |ax_1x_2 \dots x_k| \leq 1 + \sum_{i=1}^k \gamma(m_{G_i}) \\ &\leq 1 + \sum_{i=1}^k \gamma(m_G - k - 1) = 1 + k \cdot \gamma(m_G - k - 1). \end{aligned}$$

Since we know that $1 \leq k \leq m_G - 2$, this shows that

$$|w| \leq \max_{1 \leq k \leq m_G - 2} \{k \cdot \gamma(m_G - k - 1) + 1\}, \quad \text{for } m_G \geq 3,$$

which is exactly the recurrence for γ given in Definition 2.

This completes the induction proof for the upper bound.

Regarding the grammars witnessing that the bound is tight, let us consider an example, with $m = 21$. We have $\gamma(21) = 4 \cdot \gamma(16) + 1$, so the grammar G_{21} has a production of the form $A_1 \rightarrow aA_2A_2A_2A_2$, where A_1 is the start symbol, and, inductively, A_2 is the start symbol of the grammar G_{16} . More precisely, by unrolling the recurrence, we find with the aid of Table 2 that $\gamma(16) = 3 \cdot \gamma(12) + 1$, $\gamma(12) = 3 \cdot \gamma(8) + 1$, $\gamma(8) = 3 \cdot \gamma(4) + 1$, and $\gamma(4) = 2 \cdot \gamma(1) + 1$. Correspondingly, we obtain the following productions for G_{21} :

$$\begin{aligned} A_1 &\rightarrow aA_2A_2A_2A_2 \\ A_2 &\rightarrow aA_3A_3A_3 \\ A_3 &\rightarrow aA_4A_4A_4 \end{aligned}$$

$$A_4 \rightarrow aA_5A_5A_5$$

$$A_5 \rightarrow aA_6A_6$$

$$A_6 \rightarrow a,$$

with A_1 being the start symbol. The pattern is easily seen to generalize, hence the proof is completed. $\square$

Does the recurrence for $\gamma(n)$ have a closed form? The recurrence seems rather odd; the first few values are computed in Table 2. Yet, more extensive computer

n	$\{1 \cdot \gamma(n-2) + 1, 2 \cdot \gamma(n-3) + 1, \dots, (n-2) \cdot \gamma(1) + 1\}$	$\gamma(n)$
1	—	1
2	—	1
3	{2}	2
4	{2, 3}	3
5	{3, 3, 4}	4
6	{4, 5, 4, 5}	5
7	{5, 7, 7, 5, 6}	7
8	{6, 9, 10, 9, 6, 7}	10
9	{8, 11, 13, 13, 11, 7, 8}	13
10	{11, 15, 16, 17, 16, 13, 8, 9}	17
11	{14, 21, 22, 21, 21, 19, 15, 9, 10}	22
12	{18, 27, 31, 29, 26, 25, 22, 17, 10, 11}	31
13	{23, 35, 40, 41, 36, 31, 29, 25, 19, 11, 12}	41
14	{32, 45, 52, 53, 51, 43, 36, 33, 28, 21, 12, 13}	53
15	{42, 63, 67, 69, 66, 61, 50, 41, 37, 31, 23, 13, 14}	69
16	{54, 83, 94, 89, 86, 79, 71, 57, 46, 41, 34, 25, 14, 15}	94
17	{70, 107, 124, 125, 111, 103, 92, 81, 64, 51, 45, 37, 27, 15, 16}	125
18	{95, 139, 160, 165, 156, 133, 120, 105, 91, 71, 56, 49, 40, 29, 16, 17}	165
19	{126, 189, 208, 213, 206, 187, 155, 137, 118, 101, 78, 61, 53, 43, 31, 17, 18}	213
20	{166, 251, 283, 277, 266, 247, 218, 177, 154, 131, 111, 85, 66, 57, 46, 33, 18, 19}	283
21	{214, 331, 376, 377, 346, 319, 288, 249, 199, 171, 144, 121, 92, 71, 61, 49, 35, 19, 20}	377

Table 2 A few γ -values for $n = 1, 2, \dots, 21$. The values printed in boldface are the maximum among the numbers in the corresponding set.

calculations suggest that for larger n , the values of γ satisfy a certain affine integer recurrence. Computational experiments motivate the following definition.

Definition 3 For integer $n \geq 21$, let the function $\hat{\gamma}(n)$ be given by the recurrence

$$\hat{\gamma}(n) = 4 \cdot \hat{\gamma}(n-5) + 1$$

with base values $\hat{\gamma}(16) = 94$, $\hat{\gamma}(17) = 125$, $\hat{\gamma}(18) = 165$, $\hat{\gamma}(19) = 213$, and $\hat{\gamma}(20) = 283$.

In effect, the above definition amounts to five non-interfering affine recurrences, one for each remainder modulo 5. Our goal will be to prove that $\gamma(n) = \hat{\gamma}(n)$, for $n \geq 16$. The recurrence relation for $\hat{\gamma}$ is affine, so it will be easier to analyze the latter. Let us start with an easy observation.

Lemma 5 For integers r, t with $0 \leq r \leq 4$ and $t \geq 0$ holds

$$\widehat{\gamma}(16 + 5t + r) = \frac{A_r 4^t - 1}{3},$$

with $A_r = 3 \cdot \widehat{\gamma}(16 + r) + 1$. In particular, $A_0 = 283$, $A_1 = 376$, $A_2 = 496$, $A_3 = 640$, and $A_4 = 850$.

Proof By induction on t . For the base case $t = 0$, it is readily verified that $A_r = 3 \cdot \widehat{\gamma}(16 + r) + 1$, for $r = 0, 1, \dots, 4$. That is,

$$\begin{aligned} 3 \cdot \widehat{\gamma}(16) + 1 &= 3 \cdot 94 + 1 = 283, \\ 3 \cdot \widehat{\gamma}(17) + 1 &= 3 \cdot 125 + 1 = 376, \\ 3 \cdot \widehat{\gamma}(18) + 1 &= 3 \cdot 165 + 1 = 496, \\ 3 \cdot \widehat{\gamma}(19) + 1 &= 3 \cdot 213 + 1 = 640, \text{ and} \\ 3 \cdot \widehat{\gamma}(20) + 1 &= 3 \cdot 283 + 1 = 850. \end{aligned}$$

For the induction step, assume the statement holds for all values up to t . By the recursive definition of $\widehat{\gamma}$,

$$\begin{aligned} \widehat{\gamma}(16 + 5(t + 1) + r) &= 1 + 4 \cdot \widehat{\gamma}(16 + 5t + r) \\ &= 1 + 4 \left(\frac{A_r 4^t - 1}{3} \right) \\ &= 1 + \frac{A_r 4^{t+1} - 1}{3} - \frac{3}{3} \\ &= \frac{A_r 4^{t+1} - 1}{3}, \end{aligned}$$

which completes the induction step. $\square$

Our next goal is to prove, for each fixed $n \geq 21$, that the finite sequence

$$1 \cdot \widehat{\gamma}(n - 2), 2 \cdot \widehat{\gamma}(n - 3), 3 \cdot \widehat{\gamma}(n - 4), 4 \cdot \widehat{\gamma}(n - 5), \dots, (n - 17) \cdot \widehat{\gamma}(n - (n - 16)) \quad (1)$$

is unimodal,¹ with its maximum at $4 \cdot \widehat{\gamma}(n - 5)$. In a first step, we compare two items in this sequence.

Lemma 6 Let n be an integer with $n \geq 16$. Then $4 \cdot \widehat{\gamma}(n + 1) > 5 \cdot \widehat{\gamma}(n)$.

Proof We write $n = 16 + 5t + r$ with integers r, t , such that $0 \leq r \leq 4$ and $t \geq 0$. It suffices to prove that the ratio $\frac{\widehat{\gamma}(16 + 5t + r)}{\widehat{\gamma}(16 + 5t + r - 1)}$ is greater than $\frac{5}{4}$. We first consider the case $r \neq 4$. Then

$$\frac{\widehat{\gamma}(16 + 5t + r + 1)}{\widehat{\gamma}(16 + 5t + r)} = \frac{A_{r+1} 4^t - 1}{A_r 4^t - 1}.$$

¹A finite sequence $c_0, c_1, \dots, c_n$ is unimodal if there exist indices r and s , such that $c_0 \leq c_1 \leq c_2 \leq \dots \leq c_r = c_{r+1} = \dots = c_{r+s} \geq c_{r+s+1} \geq \dots \geq c_n$. In other words, sequence is unimodal if its entries rise to a maximum and then decrease.

The remaining case is $r = 4$. There

$$\frac{\widehat{\gamma}(16 + 5t + r + 1)}{\widehat{\gamma}(16 + 5t + r)} = \frac{4A_0 4^t - 1}{A_4 4^t - 1}.$$

In all of the above cases, we want to prove that this ratio is greater than $\frac{5}{4}$. This motivates us to put forward the following constants: for $0 \leq r \leq 3$, define $a_r = A_{r+1}$ and $b_r = A_r$, as well as $a_4 = 4A_0$ and $b_4 = A_4$.

Elementary analysis shows that for real constants $a > b$, the real function $f(t) = \frac{a4^t - 1}{b4^t - 1}$ is strictly decreasing with t (we have $\frac{df(t)}{dt} = \frac{(b-a)4^t \ln 4}{(b4^t - 1)^2} < 0$), and that towards infinity, we have $\lim_{t \rightarrow \infty} f(t) = \frac{a}{b}$. So

$$\frac{a_r 4^0 - 1}{b_r 4^0 - 1} > \frac{a_r 4^1 - 1}{b_r 4^1 - 1} > \frac{a_r 4^2 - 1}{b_r 4^2 - 1} > \dots > \frac{a_r}{b_r}.$$

We calculate the last ratio in the above chain of inequalities for each r :

$$\begin{aligned} r = 0 : & \quad \frac{a_0}{b_0} = \frac{125}{94} \doteq 1.32979 \\ r = 1 : & \quad \frac{a_1}{b_1} = \frac{165}{125} = 1.32 \\ r = 2 : & \quad \frac{a_2}{b_2} = \frac{213}{165} \doteq 1.29091 \\ r = 3 : & \quad \frac{a_3}{b_3} = \frac{283}{213} \doteq 1.32864 \\ r = 4 : & \quad \frac{a_4}{b_4} = \frac{376}{283} \doteq 1.32862. \end{aligned}$$

In each of the above cases, the ratio is larger than $\frac{5}{4}$, and this completes the proof. $\square$

The following corollary is almost immediate:

Corollary 7 Let n, p be integers with $n \geq 16$ and $p \geq 4$. Then

$$p \cdot \widehat{\gamma}(n+1) > (p+1) \cdot \widehat{\gamma}(n).$$

Proof For $p = 4$ we proved the statement in the previous lemma, that is,

$$\frac{\widehat{\gamma}(n+1)}{\widehat{\gamma}(n)} > \frac{5}{4}.$$

For $p > 4$, notice that we have $\frac{5}{4} > \frac{p+1}{p}$, so the statement *a fortiori* holds. $\square$

This establishes the following chain of inequalities:

$$4 \cdot \widehat{\gamma}(n-5) > 5 \cdot \widehat{\gamma}(n-6) > \dots > (n-17) \cdot \widehat{\gamma}(n-(n-16)).$$

It remains to consider the opposite end of sequence (1): In order to prove unimodality, it remains to show that

$$1 \cdot \widehat{\gamma}(n-2) < 2 \cdot \widehat{\gamma}(n-3) < \dots < 4 \cdot \widehat{\gamma}(n-5);$$

this will be done in a similar fashion.

Lemma 8 Let n be an integer with $n \geq 16$. Then $4 \cdot \hat{\gamma}(n) \geq 3 \cdot \hat{\gamma}(n+1)$.

Proof We again write $n = 16 + 5t + r$ with integers r, t , such that $0 \leq r \leq 4$ and $t \geq 0$. Slightly different to before, we now aim to prove that the ratio $\frac{\hat{\gamma}(16+5t+r+1)}{\hat{\gamma}(16+5t+r)}$ is at most $\frac{4}{3}$. Again, for $0 \leq r \leq 3$, we use the notation $a_r = A_{r+1}$ and $b_r = A_r$, as well as $a_4 = 4A_0$ and $b_4 = A_4$. Recall the chain of inequalities $\frac{a_r 4^0 - 1}{b_r 4^0 - 1} > \frac{a_r 4^1 - 1}{b_r 4^1 - 1} > \frac{a_r 4^2 - 1}{b_r 4^2 - 1} > \dots > \frac{a_r 4^t - 1}{b_r 4^t - 1} > \dots$. This time, we calculate the *first* ratio in the above chain of inequalities for each r :

$$\begin{aligned} r = 0 : \quad & \frac{a_0 - 1}{b_0 - 1} = \frac{125 - 1}{94 - 1} = \frac{4}{3} \\ r = 1 : \quad & \frac{a_1 - 1}{b_1 - 1} = \frac{165 - 1}{125 - 1} = \frac{41}{31} \doteq 1.3226 \\ r = 2 : \quad & \frac{a_2 - 1}{b_2 - 1} = \frac{213 - 1}{165 - 1} = \frac{53}{41} \doteq 1.2927 \\ r = 3 : \quad & \frac{a_3 - 1}{b_3 - 1} = \frac{283 - 1}{213 - 1} = \frac{141}{106} \doteq 1.3302 \\ r = 4 : \quad & \frac{a_4 - 1}{b_4 - 1} = \frac{376 - 1}{283 - 1} = \frac{125}{94} \doteq 1.3298 \end{aligned}$$

In each of the above cases, the ratio is *at most* $\frac{4}{3}$, and this completes the proof. $\square$

Similarly as before, we obtain a corollary for other items in the sequence:

Corollary 9 Let n, p be integers with $n \geq 16$ and $2 \leq p \leq 4$. Then

$$(p-1) \cdot \hat{\gamma}(n+1) \leq p \cdot \hat{\gamma}(n).$$

$\square$

As a synopsis of the above, we obtain the following statement.

Lemma 10 Let $n \geq 21$. Then $\hat{\gamma}(n) = \max_{1 \leq k \leq n-17} \{k \cdot \hat{\gamma}(n-k-1) + 1\}$.

Proof First consider the range where $k \leq 4$. Applying Corollary 9 three times yields

$$1 \cdot \hat{\gamma}(n-2) \leq 2 \cdot \hat{\gamma}(n-3) \leq 3 \cdot \hat{\gamma}(n-4) \leq 4 \cdot \underbrace{\hat{\gamma}(n-5)}.$$

Notice that, because we assumed $n \geq 21$, Corollary 9 applies for all three inequalities.

Complementing this, let us look at the range where $k \geq 4$ —this range has more than one element only for $n \geq 22$. By applying Corollary 7 repeatedly (a total of $n-21$ times), we obtain

$$\underbrace{4 \cdot \hat{\gamma}(n-5) > 5 \cdot \hat{\gamma}(n-6) > \dots > (n-18) \cdot \hat{\gamma}(17) > (n-17) \cdot \hat{\gamma}(16)}_{p=4, m=n-6}.$$

Also here, Corollary 7 is applicable in each case because $n \geq 22$.

So the maximum on the right-hand side of the equation is attained at $k=4$, and, using the recursive definition of $\hat{\gamma}$, we obtain $\hat{\gamma}(n) = 1 + 4 \cdot \hat{\gamma}(n-5)$ for $n \geq 21$ as desired. $\square$

Now let us extend the domain of $\hat{\gamma}$ to the set of positive integers, by defining $\hat{\gamma}(n) = \gamma(n)$ for $1 \leq n \leq 15$. Then the following is immediate:

Lemma 11 The function $\hat{\gamma}$ satisfies the recurrence

$$\hat{\gamma}(n) = \max_{1 \leq k \leq n-2} \{k \cdot \hat{\gamma}(n-k-1) + 1\}, \quad \text{for } 3 \leq n \leq 15,$$

and $\hat{\gamma}(1) = \hat{\gamma}(2) = 1$.

If we were able to demonstrate that the whole sequence

$$1 \cdot \hat{\gamma}(n-2), 2 \cdot \hat{\gamma}(n-3), \dots, (n-3) \cdot \hat{\gamma}(2), (n-2) \cdot \hat{\gamma}(1) \quad (2)$$

is unimodal, then this would show that the maximum is always attained by $4 \cdot \hat{\gamma}(n-5)$. It remains to study the last 15 items in the sequence, namely

$$(n-16) \cdot \hat{\gamma}(15), (n-15) \cdot \hat{\gamma}(14), \dots, (n-3) \cdot \hat{\gamma}(2), (n-2) \cdot \hat{\gamma}(1). \quad (3)$$

Unfortunately, that sequence is not decreasing: the last two entries $(n-3) \cdot \hat{\gamma}(2)$ and $(n-2) \cdot \hat{\gamma}(1)$ satisfy $(n-3) \cdot \hat{\gamma}(2) < (n-2) \cdot \hat{\gamma}(1)$ for all n , since $\hat{\gamma}(1) = \hat{\gamma}(2) = 1$. Nevertheless, it is easy to see that

$$(n-4) \cdot \hat{\gamma}(3) \geq (n-2) \cdot \hat{\gamma}(1), \quad \text{for } n \geq 6,$$

so it suffices to prove that the sequence shortened by its last element is in fact unimodal. By our previous investigation, it suffices to prove the following lemma:

Lemma 12 Let $n \geq 21$. Then $(n-k-2) \cdot \hat{\gamma}(k+1) \geq (n-k-1) \cdot \hat{\gamma}(k)$, for $2 \leq k \leq 15$.

Proof The inequality $(n-k-2) \cdot \hat{\gamma}(k+1) \geq (n-k-1) \cdot \hat{\gamma}(k)$ is equivalent to

$$\frac{\hat{\gamma}(k+1)}{\hat{\gamma}(k)} \geq \frac{n-k-1}{n-k-2},$$

since $\hat{\gamma}(k) > 0$ for k satisfying $2 \leq k \leq 15$.

For fixed k the right-hand side

$$\frac{n-k-1}{n-k-2} = 1 + \frac{1}{n-k-2}$$

is strictly decreasing in n . Therefore, for all $n \geq 21$ it suffices to check the worst case $n = 21$. So a sufficient condition is

$$\frac{\hat{\gamma}(k+1)}{\hat{\gamma}(k)} \geq 1 + \frac{1}{19-k},$$

for each k with $2 \leq k \leq 15$:

$$k = 2 : \quad \frac{\hat{\gamma}(3)}{\hat{\gamma}(2)} = \frac{2}{1} = 2 > \frac{18}{17} = 1 + \frac{1}{19-2}$$

$$\begin{aligned}
k=3: \quad & \frac{\widehat{\gamma}(4)}{\widehat{\gamma}(3)} = \frac{3}{2} = \frac{24}{16} > \frac{17}{16} = 1 + \frac{1}{19-3} \\
k=4: \quad & \frac{\widehat{\gamma}(5)}{\widehat{\gamma}(4)} = \frac{4}{3} = \frac{20}{15} > \frac{16}{15} = 1 + \frac{1}{19-4} \\
k=5: \quad & \frac{\widehat{\gamma}(6)}{\widehat{\gamma}(5)} = \frac{5}{4} = \frac{35}{28} > \frac{30}{28} = \frac{15}{14} = 1 + \frac{1}{19-5} \\
k=6: \quad & \frac{\widehat{\gamma}(7)}{\widehat{\gamma}(6)} = \frac{7}{5} = \frac{92}{65} > \frac{70}{65} = \frac{14}{13} = 1 + \frac{1}{19-6} \\
k=7: \quad & \frac{\widehat{\gamma}(8)}{\widehat{\gamma}(7)} = \frac{10}{7} = \frac{120}{84} > \frac{91}{84} = \frac{13}{12} = 1 + \frac{1}{19-7} \\
k=8: \quad & \frac{\widehat{\gamma}(9)}{\widehat{\gamma}(8)} = \frac{13}{10} = \frac{141}{110} > \frac{12}{110} = \frac{12}{11} = 1 + \frac{1}{19-8} \\
k=9: \quad & \frac{\widehat{\gamma}(10)}{\widehat{\gamma}(9)} = \frac{17}{13} = \frac{179}{130} > \frac{143}{130} = \frac{11}{10} = 1 + \frac{1}{19-9} \\
k=10: \quad & \frac{\widehat{\gamma}(11)}{\widehat{\gamma}(10)} = \frac{22}{17} = \frac{198}{153} > \frac{170}{153} = \frac{10}{9} = 1 + \frac{1}{19-10} \\
k=11: \quad & \frac{\widehat{\gamma}(12)}{\widehat{\gamma}(11)} = \frac{31}{22} = \frac{124}{88} > \frac{99}{88} = \frac{9}{8} = 1 + \frac{1}{19-11} \\
k=12: \quad & \frac{\widehat{\gamma}(13)}{\widehat{\gamma}(12)} = \frac{41}{31} = \frac{287}{217} > \frac{248}{217} = \frac{8}{7} = 1 + \frac{1}{19-12} \\
k=13: \quad & \frac{\widehat{\gamma}(14)}{\widehat{\gamma}(13)} = \frac{53}{41} = \frac{318}{246} > \frac{287}{246} = \frac{7}{6} = 1 + \frac{1}{19-13} \\
k=14: \quad & \frac{\widehat{\gamma}(15)}{\widehat{\gamma}(14)} = \frac{69}{53} = \frac{345}{265} > \frac{318}{265} = \frac{6}{5} = 1 + \frac{1}{19-14} \\
k=15: \quad & \frac{\widehat{\gamma}(16)}{\widehat{\gamma}(15)} = \frac{94}{69} = \frac{376}{276} > \frac{345}{276} = \frac{5}{4} = 1 + \frac{1}{19-15}
\end{aligned}$$

This proves the stated claim. $\square$

Putting everything together, we arrive at a closed form for $\gamma(n)$.

Lemma 13 Let n be a positive integer. Then

$$\gamma(n) = \begin{cases} n-1, & \text{if } 2 \leq n \leq 6, \\ \frac{1}{3}(31 \cdot 4^{\frac{n-8}{5}} - 1), & \text{if } n \bmod 5 \equiv 3 \text{ and } n \geq 8, \\ \frac{1}{3}(40 \cdot 4^{\frac{n-9}{5}} - 1), & \text{if } n \bmod 5 \equiv 4 \text{ and } n \geq 9, \\ \frac{1}{3}(52 \cdot 4^{\frac{n-10}{5}} - 1), & \text{if } n \bmod 5 \equiv 0 \text{ and } n \geq 10, \\ \frac{1}{3}(94 \cdot 4^{\frac{n-12}{5}} - 1), & \text{if } n \bmod 5 \equiv 2 \text{ and } n \geq 12, \\ \frac{1}{3}(283 \cdot 4^{\frac{n-16}{5}} - 1), & \text{if } n \bmod 5 \equiv 1 \text{ and } n \geq 16. \end{cases}$$

Also, $\gamma(1) = 1$, $\gamma(7) = 7$, $\gamma(11) = 22$, $\gamma(13) = 41$, $\gamma(14) = 53$, and $\gamma(15) = 69$.

Proof We demonstrate first that $\widehat{\gamma}(n) = \gamma(n)$, for $n \geq 1$. Comparing Definition 2 and Lemma 11, we see that this is the case for $n \leq 15$.

Assume next that $n \geq 21$. Here we have $\hat{\gamma}(n) = \max_{1 \leq k \leq n-17} \{k \cdot \hat{\gamma}(n-k-1) + 1\}$ by Lemma 10; and Lemma 12 yields the chain of inequalities

$$(n-17) \cdot \hat{\gamma}(16) \geq (n-16) \cdot \hat{\gamma}(n-15) \geq \cdots (n-4) \hat{\gamma}(3) \geq (n-3) \cdot \hat{\gamma}(2).$$

Also, we have $(n-4) \hat{\gamma}(3) \geq (n-2) \hat{\gamma}(1)$ for $n \geq 6$. We can thus extend the range of the maximum and obtain $\hat{\gamma}(n) = \max_{1 \leq k \leq n-2} \{k \cdot \hat{\gamma}(n-k-1) + 1\}$. This coincides with the equation in Definition 2.

For $16 \leq n \leq 20$, it remains to look up the values $\gamma(n)$ in Table 2 for each n , and find that the values coincide with the base values of the recurrence for $\hat{\gamma}(n)$ as given in Definition 3.

Regarding the closed form, we shall present the details for the case $\frac{1}{3}(31 \cdot 4^{\frac{n-8}{5}} - 1)$, if $n \bmod 5 \equiv 3$ and $n \geq 8$ and leave the rest to the reader. For $n \in \{8, 13\}$, we can verify equality using Table 2. For $n \geq 23$, we can write $n = 18 + 5t$ with $t \geq 0$, and then use Lemma 5 to obtain

$$\begin{aligned} \hat{\gamma}(18 + 5t) &= \frac{A_2 \cdot 4^t - 1}{3} \\ &= \frac{31 \cdot 4^2 \cdot 4^t - 1}{3} \\ &= \frac{31 \cdot 4^{t+2} - 1}{3} \\ &= \frac{31 \cdot 4^{\frac{n-8}{5}} - 1}{3}, \end{aligned}$$

where for the last line, we used the equality $n - 8 = 5t + 10$. This establishes equality for the case $n \bmod 5 \equiv 3$ and $n \geq 8$, as desired. $\square$

We are finally in position to state our main result—it immediately follows from Theorem 4 and Lemma 13.

Theorem 14 *Let L be generated by the context-free grammar $G = (N, T, P, S)$ in Greibach normal form, where L is not empty. Then*

$$\text{mpcf}(L) \leq \gamma(m_G) + 1,$$

where $m_G := \sum_{(A \rightarrow \alpha) \in P} |\alpha|$ and γ is the function given by

$$\gamma(n) = \begin{cases} n-1, & \text{if } 2 \leq n \leq 6, \\ \frac{1}{3}(31 \cdot 4^{\frac{n-8}{5}} - 1), & \text{if } n \bmod 5 \equiv 3 \text{ and } n \geq 8, \\ \frac{1}{3}(40 \cdot 4^{\frac{n-9}{5}} - 1), & \text{if } n \bmod 5 \equiv 4 \text{ and } n \geq 9, \\ \frac{1}{3}(52 \cdot 4^{\frac{n-10}{5}} - 1), & \text{if } n \bmod 5 \equiv 0 \text{ and } n \geq 10, \\ \frac{1}{3}(94 \cdot 4^{\frac{n-12}{5}} - 1), & \text{if } n \bmod 5 \equiv 2 \text{ and } n \geq 12, \\ \frac{1}{3}(283 \cdot 4^{\frac{n-16}{5}} - 1), & \text{if } n \bmod 5 \equiv 1 \text{ and } n \geq 16. \end{cases}$$

Also, $\gamma(1) = 1$, $\gamma(7) = 7$, $\gamma(11) = 22$, $\gamma(13) = 41$, $\gamma(14) = 53$, and $\gamma(15) = 69$. Furthermore, for every $m \geq 1$, there exists a context-free grammar G_m witnessing that this bound is tight.

5 Conclusion

In this work, we analyzed the relation between the pumping constant and the size of a smallest grammar in Greibach normal form of a given context-free language. The

worst case is attained by a singleton unary language; the grammars witnessing the optimal bound $\gamma(n)$ are based on (eventually) recursively quadrupling the length of the string—along with producing a single terminal symbol with each rule, of course. Interestingly, there is a transient phase for small n (up to 20), where the optimal strategy is not quadrupling, but tripling or doubling. Also, the growth of $\gamma(n)$ depends on the remainder of n modulo 5. For grammars in 2-standard form, the optimal bound $\gamma_2(n)$ is simpler. Again, the growth rate is not uniform, but depends on the remainder of n modulo 2.

For unrestricted context-free grammars, the corresponding question was studied recently in [7]. There, the optimal grammars are also generating a single unary word, but are based on recursively tripling the length of the string, instead of quadrupling. The function $\rho(n)$ describing the optimal bound in the unrestricted case is much simpler: it is the Moon-Moser [12] sequence, with $\rho(n) = 3^{n/3}$ if n is divisible by 3, and with similar formulas for each of the other two remainders modulo 3. Ordering the obtained bounds by growth rate, we have $\gamma_2(n)$, $\gamma(n)$, and $\rho(n)$ in increasing order. This nicely fits with the intuition that imposing structural constraints on the grammar sacrifices succinctness.

The integer recurrence $\gamma(n) = \max_{1 \leq k \leq n-2} \{k \cdot \gamma(n-k-1) + 1\}$ nicely combines the basic operations increment, multiplication and maximum on integers. It can be seen as an affine variant on the keystroke problem studied in [15], and had not yet been covered by the On-line Encyclopedia of Integer Sequences. As a result of this work, the sequence $(\gamma(n))_{n \geq 1}$ was newly added as A389642. The techniques used in this paper may also be helpful in solving other non-convex recurrences on integers.

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